

Evaluation of Point-Cloud Segmentation Approaches for Detecting Walkable Steps on Staircases in the Presence of Sensor Noise

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Abstract—Reliable three-dimensional environmental perception is essential for stair-climbing robots, particularly to differentiate safely traversable horizontal surfaces from potentially dangerous structural components. This paper presents a segmentation pipeline applied to a synthetic point cloud of a four-step staircase, where the geometry is derived from mathematical equations and augmented with simulated LiDAR noise and outliers. Four distinct segmentation strategies are developed and benchmarked: RANSAC-based plane fitting, PCA-driven normal vector classification, DBSCAN clustering combined with slope analysis, and height histogram-based segmentation. The primary contribution is a controlled, fully annotated synthetic staircase dataset with adjustable noise, together with a unified evaluation framework and a systematic investigation of sensor-noise sensitivity from the perspective of robot safety. Walkable-class performance is assessed using accuracy, precision, recall, F1 score, and intersection over union (IoU). Among the evaluated approaches, the height histogram and normal vector methods deliver the strongest results, with F1 scores of 0.883 and 0.876 respectively, and precision values exceeding 0.93. A noise sensitivity study reveals that all metrics decline monotonically with increasing noise: the F1 score falls from 0.970 under ideal conditions to 0.329 when the noise standard deviation reaches 0.08 m. The safety analysis concludes that while these segmentation approaches are capable of forming a first-layer perception module, they must be combined with geometric safety margins and sensor fusion before being deployed on an autonomous robot.

Index Terms—point cloud, segmentation, RANSAC, normal vector, DBSCAN, stair detection, robot

I. INTRODUCTION

Enabling mobile robots to ascend and descend staircases remains an open challenge in robotics, since staircase structures are ubiquitous in indoor human environments. Such robots must reliably distinguish the horizontal tread surfaces (where the robot can place its feet or wheels) from vertical riser surfaces and from the potentially hazardous step edges. Misclassification of any of these regions can result in slipping or falling, making segmentation accuracy a direct safety requirement.

Modern range sensors such as LiDAR units and depth cameras deliver the environment as three-dimensional point clouds. Segmenting point clouds to recover geometric primitives like staircase components demands techniques that can separate

points according to surface orientation and elevation. Classical approaches include RANSAC-based plane estimation [1], [2], surface normal computation [5], [6], and density-based spatial clustering [3], [4]. These foundations have been applied to staircase detection from 3-D point clouds and depth imagery, in contexts ranging from mobility aids to autonomous mobile robots [9]–[12].

The goals of this study are fourfold: (i) to produce a simulated point cloud of a four-step staircase using closed-form geometric equations augmented by sensor noise; (ii) to implement and compare several classical segmentation algorithms for tread, riser, and other-class separation; (iii) to quantify method performance and characterise the effect of noise; and (iv) to evaluate the readiness of each method for deployment on a real stair-climbing robot.

The contributions can be summarised as: (a) a reproducible, fully labeled synthetic benchmark point cloud with tunable noise and outlier fraction; (b) a unified, side-by-side comparison of four classical segmentation methods under identical experimental conditions; and (c) a noise sensitivity study paired with a safety-oriented analysis of the precision levels required for safe robot operation on stairs.

II. POINT CLOUD GENERATION

This section describes the construction of the synthetic staircase point cloud. The staircase is modelled with four steps whose geometric parameters are listed in Table I. Every generated point carries a ground-truth class label drawn from three categories: *tread* (walkable horizontal surface), *riser* (vertical connecting surface), and *other* (covering noise points and outliers).

A. Horizontal Surface (Tread)

The tread is the flat, horizontal area upon which the robot physically rests during stair traversal. For step index $i \in \{0, 1, 2, 3\}$, sample coordinates are drawn according to

$$\begin{cases} x \sim \text{Uniform}(0, w) \\ y \sim \text{Uniform}(i d, (i + 1) d) \\ z = i h + \varepsilon_z, \quad \varepsilon_z \sim \mathcal{N}(0, \sigma_z^2) \end{cases} \quad (1)$$

TABLE I
STAIRCASE GEOMETRIC PARAMETERS

Parameter	Symbol	Value
Staircase width (x -axis)	w	2.5 m
Tread depth (y -axis)	d	0.30 m
Step height (z -axis)	h	0.18 m
Number of steps	n	4
Vertical noise std. deviation	σ_z	0.02 m
LiDAR noise std. deviation	σ_L	0.005 m
Outlier fraction	—	4%

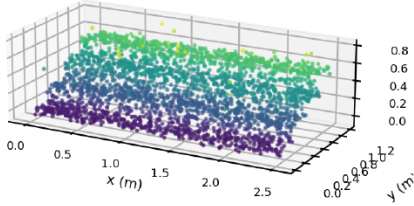


Fig. 1. Synthetic point cloud of the four-step staircase. Colour encodes height z ; the four distinct step levels are clearly visible alongside the scattered outlier points.

The additive term ε_z models measurement uncertainty in the vertical direction, so the tread surface deviates slightly from a perfectly flat plane.

B. Vertical Surface (Riser)

The riser connects two consecutive tread levels and is oriented perpendicular to the ground plane. Points on riser i with $i \in \{1, 2, 3\}$ are generated as

$$\begin{cases} x \sim \text{Uniform}(0, w) \\ y = i d + \varepsilon_y, \quad \varepsilon_y \sim \mathcal{N}(0, \sigma_y^2) \\ z \sim \text{Uniform}((i-1)h, ih) \end{cases} \quad (2)$$

The riser at $i = 0$ is omitted because the corresponding height range falls below floor level, which is geometrically inconsistent with an ascending staircase.

C. LiDAR Noise and Outliers

After the structured tread and riser points are created, isotropic Gaussian noise $\mathcal{N}(0, \sigma_L^2)$ is applied independently along all three coordinate axes to replicate the measurement uncertainty of a real LiDAR sensor. An additional four percent of the total point count is then inserted as uniformly distributed random outliers scattered throughout the scene bounding box. The resulting cloud contains 15 808 points in total: 10 400 tread points, 4 800 riser points, and 608 outlier or noise points. A visualisation of the synthetic point cloud is shown in Fig. 1.

III. SEGMENTATION METHODOLOGY

Four segmentation strategies are developed and evaluated. Because most of them rely on locally estimated surface normals, normal vector computation is described first.

A. Normal Vector Estimation

At each point, the surface normal is recovered through Principal Component Analysis (PCA) applied to the k -nearest neighbourhood, a widely adopted technique in point-cloud processing [5]–[7]. The local covariance matrix is constructed as

$$\mathbf{C}_p = \frac{1}{k} \sum_{j=1}^k (\mathbf{q}_j - \bar{\mathbf{q}})(\mathbf{q}_j - \bar{\mathbf{q}})^\top \quad (3)$$

where $\mathbf{q}_j$ denotes a neighbouring point and $\bar{\mathbf{q}}$ is the neighbourhood centroid. The normal vector $\mathbf{n}_p$ is the eigenvector of $\mathbf{C}_p$ associated with its smallest eigenvalue:

$$\mathbf{n}_p = \arg \min_{\|\mathbf{v}\|=1} \mathbf{v}^\top \mathbf{C}_p \mathbf{v} \quad (4)$$

The magnitude of the vertical normal component, $|n_z|$, serves as an orientation indicator: values near one indicate a near-horizontal surface, whereas values near zero indicate a near-vertical surface. All experiments use $k = 30$.

B. Method 1: RANSAC Plane Fitting

RANSAC [1], [2] is employed to isolate horizontal planes whose normals are closely aligned with the z -axis. In each iteration, three points are sampled at random to define a candidate plane $\mathbf{n}^\top \mathbf{x} + d = 0$. A candidate is retained only when $|n_z| > 0.9$, and its support is measured as the count of points lying within a distance threshold $\tau = 0.02$ m of the plane. The plane with the largest inlier set is extracted iteratively through inlier removal. Because a purely geometric horizontal band may inadvertently include riser points at the same elevation, plane membership is further restricted to points whose local normal satisfies $|n_z| > 0.8$. This highlights the need to complement geometric plane fitting with normal-vector consistency checks.

C. Method 2: PCA-Based Normal Vector Analysis

Each point is classified directly from the vertical component of its estimated normal vector:

$$\text{label}(p) = \begin{cases} \text{tread} & |n_z| > \tau_h \\ \text{riser} & |n_z| < \tau_v \\ \text{other} & \text{otherwise} \end{cases} \quad (5)$$

with threshold values $\tau_h = 0.85$ and $\tau_v = 0.35$. Points whose normal orientation falls in the intermediate zone are treated as edge or transition regions.

D. Method 3: DBSCAN Clustering with Slope Analysis

On a compact staircase, adjacent tread and riser surfaces share boundaries, causing pure proximity-based clustering to merge them into a single region. To address this, DBSCAN [3], [4] is applied primarily as an outlier rejection step with parameters $\varepsilon = 0.04$ m and a minimum neighbourhood size of ten points. Points flagged as noise by DBSCAN receive the *other* label, while the remaining points are classified through

TABLE II
SEGMENTATION PERFORMANCE COMPARISON FOR THE WALKABLE CLASS

Method	Acc	Prec	Rec	F1	IoU
RANSAC + Normal	0.836	0.947	0.795	0.864	0.761
PCA + Normal	0.848	0.944	0.816	0.876	0.779
DBSCAN + Slope	0.843	0.957	0.798	0.870	0.770
Height Histogram	0.854	0.935	0.836	0.883	0.790

the slope analysis defined in (5). The combination of density filtering and orientation analysis enhances robustness to outlier contamination.

E. Method 4: Height Histogram Analysis

Tread surfaces are distributed at discrete elevation levels approximating $z \approx 0, h, 2h, 3h$. A histogram of the z -coordinates therefore displays a pronounced peak at each tread level. The n highest peaks separated by at least eight centimetres are selected as tread height estimates. Points within ± 0.04 m of a detected peak and whose normal is predominantly vertical ($|n_z| > 0.75$) are labelled as tread. The detected peak heights are 0.009, 0.181, 0.352, and 0.542 m, in close agreement with the nominal integer multiples of the step height h .

IV. EXPERIMENTAL RESULTS

Segmentation quality is evaluated on the walkable (tread) class, treating tread as the positive label. From the counts of true positives (TP), false positives (FP), false negatives (FN), and true negatives (TN), five metrics are derived:

$$\begin{aligned} \text{Acc} &= \frac{TP + TN}{N}, & \text{Prec} &= \frac{TP}{TP + FP}, & \text{Rec} &= \frac{TP}{TP + FN} \\ F1 &= \frac{2 \text{Prec} \cdot \text{Rec}}{\text{Prec} + \text{Rec}}, & \text{IoU} &= \frac{TP}{TP + FP + FN} \end{aligned} \quad (6)$$

Table II presents the quantitative results. The height histogram method attains the highest F1 and IoU scores, while DBSCAN achieves the highest precision. Every method records precision above 0.93, confirming that points labelled as walkable are almost exclusively genuine treads. A graphical comparison of all metrics across methods appears in Fig. 2.

The height distribution of the point cloud along with the automatically detected tread peaks is displayed in Fig. 3. The four well-separated peaks confirm that the histogram-based approach can reliably identify individual step levels. A side-by-side comparison of ground-truth labels and the height histogram segmentation output is provided in Fig. 4.

V. ANALYSIS AND DISCUSSION

A. Sensitivity to Sensor Noise

To evaluate how noise intensity affects segmentation quality, the vertical noise standard deviation σ_z is swept from 0 to 0.08 m while re-running the normal analysis pipeline at each noise level. The resulting curves in Fig. 5 demonstrate a strictly monotonic performance degradation. The F1 score starts at 0.970 in the noise-free case, decreases to 0.875 at

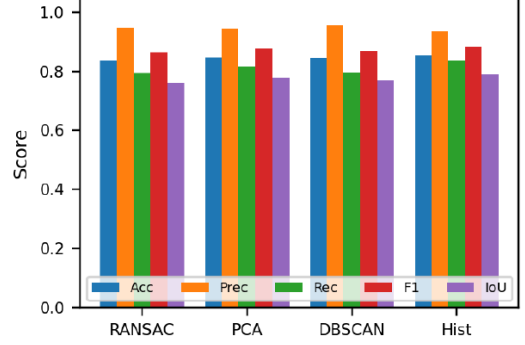


Fig. 2. Visual comparison of segmentation metrics across the four methods for the walkable class.

ogram approach effectively separates the staircase levels. The three-dimensional segmentation result compared with the ground-truth labels is shown in Fig. 4.

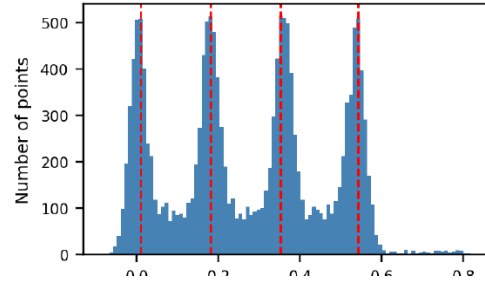


Fig. 3. Distribution of point heights along the z -axis. Dashed red vertical lines indicate the detected tread elevation levels.

$\sigma_z = 0.02$ m, then drops sharply to 0.551 at $\sigma_z = 0.04$ m, and reaches only 0.329 at $\sigma_z = 0.08$ m. This behaviour arises because elevated noise corrupts local normal vector estimates, making it impossible for fixed orientation thresholds to reliably differentiate treads from risers.

B. Implications for Stair-Climbing Robots

From a robot safety standpoint, precision is more consequential than recall in staircase segmentation. A false positive, a riser or edge misclassified as a walkable tread, poses an immediate falling hazard, whereas a false negative merely leaves a valid tread undetected. Although the ~ 0.94 precision reported here is encouraging, practical deployment demands values beyond 0.99, achievable through post-processing steps such as edge suppression, morphological erosion of the detected tread masks, and the imposition of a geometric safety margin around step boundaries.

Furthermore, the strong sensitivity to noise emphasises the importance of either high-quality sensing hardware or temporal averaging across successive scans, consistent with the approach taken in odometry-assisted stair traversal systems [11].

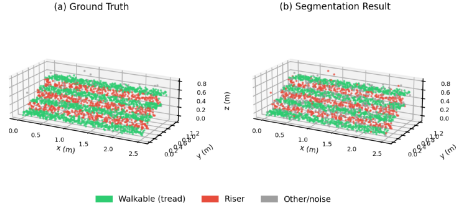


Fig. 4. Comparison of segmentation results: (a) ground-truth class labels; (b) output of the height histogram method. Green regions represent walkable treads, red regions represent risers, and grey regions correspond to the other class.

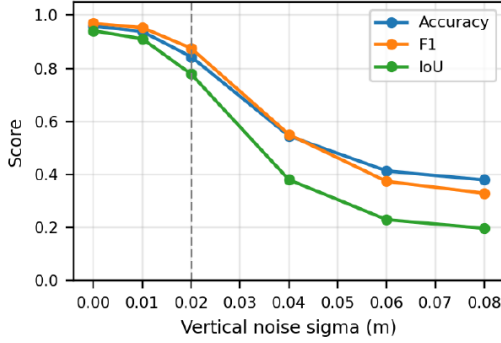


Fig. 5. Effect of vertical noise standard deviation on accuracy, F1 score, and IoU. The dashed vertical line indicates the noise level used in the main experiment ($\sigma_z = 0.02$ m).

The orientation threshold $|n_z|$ should ideally be adapted to the locally estimated noise level rather than fixed globally. Of particular concern is the step-edge region, which carries a transitional normal vector that resists clean classification; a dedicated *edge* class paired with an explicit no-step zone is therefore recommended.

Based on this analysis, the evaluated segmentation methods are adequate as a first-layer perceptual module but are insufficient as a standalone safety guarantee. Full deployment readiness requires integration with an inertial measurement unit and a visual camera, temporal verification of detections, geometric safety margins, and a contact-based fail-safe fallback.

VI. CONCLUSION

This work has presented a complete end-to-end pipeline for walkable surface segmentation on a synthetic four-step staircase point cloud. The dataset is generated entirely from analytic geometry equations, supplemented with isotropic LiDAR noise and uniform outliers, yielding more than 15 000 labeled points for controlled evaluation. Four segmentation methods were implemented and benchmarked under identical conditions: RANSAC-based plane fitting, PCA normal analysis, DBSCAN with slope filtering, and height histogram segmentation. Both the height histogram and normal analysis

approaches achieved the highest performance, with F1 scores approaching 0.88 and IoU values near 0.79. A systematic noise sensitivity study confirmed that all methods degrade monotonically with increasing sensor uncertainty, stressing the importance of sensor quality and robust preprocessing.

Future work will focus on the following directions: defining a dedicated step edge class to handle transitional surface regions; developing noise-adaptive orientation thresholds; incorporating smoothness-based region growing [8] for more spatially coherent labels; exploring learning-based detectors [12] for improved generalisation; and validating the pipeline on real-world LiDAR and depth-camera point cloud data.

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